ALBAR: MOTION-CENTRIC ADVERSARIAL DEBIASING WITH A LEAKAGE-REDUCED SCUBA/SCUFO BENCHMARK

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ABSTRACT

Action recognition models often solve the wrong problem: they exploit static appearance cues such as scenes, objects, or actor identity instead of the motion that defines the action. This shortcut leads to brittle performance under distribution shift and is hard to diagnose when evaluation protocols themselves leak contextual information. We present ALBAR, a simple end-to-end debiasing method that trains a video classifier to remain discriminative on motion clips while becoming deliberately uninformative on motionless counterparts. ALBAR pairs each training clip with a static clip formed by tiling one video frame, then combines standard classification with a static adversarial loss, entropy maximization on static predictions, and a gradient penalty for stable optimization. In parallel, we introduce a leakage-reduced UCF101-SCUBA/SCUFO benchmark that replaces coarse box-based compositing with tighter segmentation-based foreground extraction, yielding cleaner tests of background and foreground static bias. On HMDB51 SCUBA/SCUFO, ALBAR improves contrasted accuracy from 40.28% with the strongest prior baseline to 53.22%, and reaches 53.68% when combined with StillMix. On the updated UCF101 protocol, ALBAR improves contrasted accuracy from 17.53% for the baseline and 24.40% for StillMix to 26.23%, while reducing SCUFO accuracy to 0.14%. The learned representation also transfers with consistent gains to UCF_Crime anomaly detection and THUMOS14 temporal localization. These results suggest that directly suppressing static evidence, together with a cleaner benchmark, meaningfully improves motion-grounded action recognition.

1 INTRODUCTION

Modern video classifiers are highly accurate on standard benchmarks such as UCF101 and HMDB51, but their success often depends on shortcuts rather than genuine motion understanding (Carreira & Zisserman, 2017; Li et al., 2023; Geirhos et al., 2020). A model can predict *skiing* from snow, *basketball* from a court, or even infer an action from a person's appearance, while paying limited attention to the temporal evidence that should define the class. This behavior degrades robustness when backgrounds, objects, or subject appearance shift out of distribution.

Static shortcut learning in action recognition has recently been studied from both the algorithmic and evaluation perspectives. Prior methods attempt to weaken scene bias through reconstruction, contrastive learning, explicit attribute supervision, or augmentation (Wang et al., 2021; Ding et al., 2022; Duan et al., 2022; Bao et al., 2021; Zhai et al., 2023; Li et al., 2023). At the same time, SCUBA and SCUFO were introduced as controlled tests for background and foreground static bias (Li et al., 2023). However, existing protocols still leave an important gap: if the synthetic benchmark itself leaks background evidence through imperfect masks, then OOD evaluation underestimates how much the classifier is still exploiting static cues.

This paper addresses both sides of the problem. First, we propose ALBAR, a label-free adversarial training objective that suppresses confidence on motionless inputs without requiring bias annotations, external critics, or explicit scene labels. The method is simple: from each video clip, we construct a

static counterpart by repeating one frame across time. We then optimize the classifier to predict the correct label on the original clip while becoming maximally uncertain on the static clip. This targets a minimal principle: if a motionless clip remains highly class-discriminative, then the model is still using spatial shortcuts.

Second, we introduce an updated UCF101-SCUBA/SCUFO protocol designed to reduce background leakage. We refine foreground extraction with tighter segmentation-based masks and preserve the original action labels while generating controlled OOD variants that isolate background and foreground bias. The resulting benchmark is intended both as an evaluation resource and as a sharper diagnostic for future debiasing work.

Our contributions are:

- We propose ALBAR, an end-to-end debiasing objective that combines temporal classification, static adversarial supervision, entropy maximization, and gradient regularization to suppress static shortcut reliance.
- We introduce a leakage-reduced UCF101-SCUBA/SCUFO benchmark built with tighter actor segmentation and matched OOD protocols for background bias, foreground bias, and conflicting foreground cues.
- We show strong gains on HMDB51 SCUBA/SCUFO and on the updated UCF101 protocol, plus consistent improvements on downstream anomaly detection and temporal localization tasks.

2 Related Works

Action recognition and motion modeling. Video recognition has progressed rapidly with large-scale pretraining and stronger backbones, including I3D, R(2+1)D, and Video Swin Transformer (Carreira & Zisserman, 2017; Tran et al., 2018; Liu et al., 2022). These models are effective on IID benchmarks, but strong IID accuracy does not by itself guarantee that the representation is motion-centric. The gap between in-distribution performance and true temporal reasoning is especially visible on out-of-context evaluations (Weinzaepfel & Rogez, 2021) and on static-bias stress tests (Li et al., 2023).

Debiasing action representations. Recent work has shown that action models are sensitive to scene and appearance shortcuts. Augmentation-based approaches such as VideoMix, Background Erasing, FAME, and StillMix manipulate spatial evidence to reduce shortcut reliance (Yun et al., 2020; Wang et al., 2021; Ding et al., 2022; Li et al., 2023). Other approaches incorporate explicit debiasing modules or adversarial criteria, for example through multi-aspect debiasing, open-set calibration, or scene-adversarial objectives (Duan et al., 2022; Bao et al., 2021; Zhai et al., 2023). More broadly, adversarial invariance learning has been studied in fairness and debiasing settings outside video recognition (Zhang et al., 2018; Bahng et al., 2020). Our method differs in two ways: it does not require bias labels or a separate bias classifier, and it directly uses the model's behavior on motionless clips as the signal to suppress.

Benchmarks for shortcut sensitivity. Benchmark design is central when the goal is to measure reliance on spurious evidence. SCUBA and SCUFO provide a clean idea: alter backgrounds or remove motion while preserving the action label, then evaluate whether a model remains correct for the right reason (Li et al., 2023). Related work has also emphasized rare-scene evaluation and out-of-context testing as necessary complements to IID accuracy (Duan et al., 2022; Weinzaepfel & Rogez, 2021). Our benchmark contribution is narrower but more diagnostic: we revisit UCF101-SCUBA/SCUFO to reduce foreground-mask leakage, so that high performance more reliably reflects robustness to static bias rather than residual compositing artifacts.

3 BENCHMARK DESIGN

3.1 GOAL AND TASK DEFINITION

The benchmark targets two failure modes. In **SCUBA**, the moving foreground is preserved while the background is replaced, testing whether the model over-relies on scene context. In **SCUFO**, motion is removed by repeating a single frame over time, testing whether the model can still classify actions from static foreground appearance alone. We also include a **Conflicting-FG** setting, where foreground appearance cues and motion cues disagree, to further probe whether the representation follows motion or static identity cues.

For evaluation, we report top-1 accuracy on the standard IID test split and on each OOD variant. Following the paired SCUBA/SCUFO design, we also use **contrasted accuracy**, which counts an example as correct only if the model predicts the correct label on the SCUBA clip and fails to produce the same correct prediction on its paired SCUFO clip. This metric is intentionally strict: it rewards models that succeed when motion is preserved and fail when only static evidence remains.

3.2 LEAKAGE-REDUCED UCF101-SCUBA/SCUFO

The original UCF101 SCUBA/SCUFO benchmark is conceptually useful, but coarse foreground extraction can still leak scene evidence into the composited clip. Figure 2 illustrates the problem: snow remains visible around the skier in the earlier protocol, making background bias harder to isolate. Our updated benchmark replaces box-like masks with tighter segmentation-based actor extraction, with the explicit goal of minimizing residual background cues while preserving realistic motion boundaries.

The updated protocol starts from UCF101 videos (Soomro et al., 2012). Foreground initialization uses available actor localization cues, which are then refined into frame-level masks using a SAM-based video tracking pipeline (Cheng et al., 2023; Kirillov et al., 2023). Backgrounds are sourced from Places365 scenes (Zhou et al., 2017), text-guided VQGAN-CLIP generations (Crowson et al., 2022), and synthetic sinusoidal textures. The resulting composites retain the original action label and test split, but provide cleaner separation between action foreground and non-causal background.

3.3 CONSTRUCTION PROTOCOL

For each source video, SCUBA is generated by compositing the original moving foreground onto a novel background. SCUFO is then generated by taking a single composited frame and stacking it across time, so that the spatial evidence remains but motion is removed. This pairing keeps labels aligned across SCUBA and SCUFO and makes contrasted accuracy well defined. It also supports per-video diagnosis: if a model succeeds on SCUBA yet remains correct on the paired SCUFO clip, the remaining signal is static rather than temporal. We preserve source-video identity across paired variants and evaluate each action under natural scenes, generated scenes, and sinusoidal textures so that semantic, low-level, and compositing-sensitive shortcuts are all exposed. We manually inspect the test clips to remove obvious compositing failures and ensure that the benchmark measures the intended bias rather than synthetic artifacts.

4 METHOD

4.1 SETUP

Let f_θ denote the action classifier and let $\boldsymbol{x} \in \mathbb{R}^{T\times H\times W\times 3}$ be a training clip with label y. In our implementation, f_θ is a Swin3D-T backbone pretrained on Kinetics-400 (Liu et al., 2022; Carreira & Zisserman, 2017). During training, we construct a static companion clip $\bar{\boldsymbol{x}}$ by selecting one frame from $\boldsymbol{x}$ and tiling it across the temporal dimension. The default choice in our code is the middle frame, which we later ablate.

The key assumption is simple: if the classifier can confidently predict the action from $\bar{\boldsymbol{x}}$, then it is using static cues that should not dominate recognition. ALBAR therefore learns from both $\boldsymbol{x}$ and $\bar{\boldsymbol{x}}$ in the same forward pass.

4.2 TRAINING OBJECTIVE

The temporal clip should remain discriminative, so we optimize the standard cross-entropy

$$\mathcal{L}_{\mathrm{CE}} = \mathrm{CE}(f_\theta(\boldsymbol{x}), y). \tag{1}$$

For the static clip, we intentionally reverse the usual supervision signal by *subtracting* the cross-entropy on $\bar{\boldsymbol{x}}$:

$$\mathcal{L}_{\mathrm{adv}} = \mathrm{CE}(f_\theta(\bar{\boldsymbol{x}}), y). \tag{2}$$

In the final loss, this term is multiplied by a negative sign, so optimizing the total objective discourages the model from preserving label information in motionless inputs. This is an adversarial objective, but unlike gradient-reversal or auxiliary-critic methods, it requires no extra network.

We further maximize predictive uncertainty on static clips. Let $p_\theta(\bar{\boldsymbol{x}}) = \mathrm{softmax}(f_\theta(\bar{\boldsymbol{x}}))$. The implementation minimizes

$$\mathcal{L}_{\mathrm{ent}} = \frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{C} p_{ic}\log p_{ic}, \tag{3}$$

which is the negative entropy; minimizing it drives the distribution toward uniformity.

Finally, we add a gradient penalty on the static branch for optimization stability:

$$\mathcal{L}_{\mathrm{gp}} = \frac{1}{B}\sum_{i=1}^{B}\left\|\nabla_{\bar{\boldsymbol{x}}_i}\sum_{c=1}^{C} f_\theta(\bar{\boldsymbol{x}}_i)_c\right\|_2, \tag{4}$$

which is inspired by gradient regularization in adversarial optimization (Gulrajani et al., 2017). The complete objective is

$$\mathcal{L} = \lambda_{\mathrm{ce}}\mathcal{L}_{\mathrm{CE}} - \lambda_{\mathrm{adv}}\mathcal{L}_{\mathrm{adv}} + \lambda_{\mathrm{ent}}\mathcal{L}_{\mathrm{ent}} + \lambda_{\mathrm{gp}}\mathcal{L}_{\mathrm{gp}}. \tag{5}$$

In the released configuration, we use $\lambda_{\mathrm{ce}} = \lambda_{\mathrm{adv}} = 1$, $\lambda_{\mathrm{ent}} = 4.25$, and $\lambda_{\mathrm{gp}} = 10.0$.

4.3 IMPLEMENTATION DETAILS

All experiments use Swin3D-T with Kinetics-400 initialization. The training clip length is 32 frames with stride 2. We train with AdamW for 50 epochs, learning rate 10^{-5}, weight decay 5×10^{-2}, random resized crop and horizontal flip, and 224×224 spatial resolution. Validation is performed every two epochs. Because ALBAR acts on the model's own static predictions, it is naturally compatible with augmentation methods; in particular, we test combination with StillMix (Li et al., 2023).

5 EXPERIMENTS

5.1 BENCHMARKS AND METRICS

We evaluate on HMDB51 (Kuehne et al., 2011) using the SCUBA/SCUFO protocols of Li et al. (2023), and on our leakage-reduced UCF101 protocol based on UCF101 (Soomro et al., 2012). Results are reported as top-1 accuracy on IID and OOD splits, with higher being better for IID, Avg SCUBA, Conflicting-FG, and contrasted accuracy, and lower being better for Avg SCUFO.

5.2 MAIN RESULTS ON HMDB51

Table 1 compares ALBAR against standard augmentation and debiasing baselines. The strongest prior method in this table is StillMix, with 40.28% contrasted accuracy. ALBAR improves this to 53.22%, a gain of 12.94 points, while also increasing Avg SCUBA from 51.81% to 53.53% and reducing Avg SCUFO from 13.39% to 1.50%. These numbers indicate that ALBAR is not merely degrading confidence globally; it preserves or improves performance when motion is available and collapses performance when only static evidence remains.

The IID cost is modest but real: 72.81% for ALBAR versus 73.92% for the unmodified baseline and 74.82% for StillMix. This tradeoff is important to report honestly. Debiasing here improves OOD reliance on motion at the expense of some standard-set accuracy. When combined with StillMix, however, the method recovers IID performance to 74.31% and further improves contrasted accuracy to 53.68%, showing that ALBAR is complementary to augmentation-based debiasing rather than a replacement for it.

5.3 Results on the updated UCF101 benchmark

Table 2 reports results on the leakage-reduced UCF101 protocol. Because this benchmark is newly constructed, we compare against the baseline model and StillMix re-evaluated under the updated masking procedure. ALBAR improves contrasted accuracy from 17.53% to 26.23%, improves Avg SCUBA from 18.78% to 26.25%, and reduces Avg SCUFO from 1.50% to 0.14%. Conflicting-FG accuracy also rises from 49.36% to 54.02%.

The updated benchmark therefore remains challenging even for strong IID models: the baseline reaches 95.51% IID accuracy, but only 17.53% contrasted accuracy. This gap reinforces the need for a dedicated OOD protocol. When combined with StillMix, ALBAR reaches 31.14% contrasted accuracy, the best result in the table.

5.4 Downstream transfer

Debiasing is most useful if it improves the representation beyond the synthetic benchmark itself. Table 3 evaluates the HMDB51-debiased model on two downstream tasks: anomaly detection on UCF_Crime (Sultani et al., 2018) and temporal action localization on THUMOS14 (Jiang et al., 2014). The gains are consistent but smaller than on SCUBA/SCUFO: anomaly detection AUC improves from 82.39 to 84.91, and THUMOS14 performance improves from 54.89 to 55.20. We view this result as encouraging but measured: ALBAR clearly helps robustness on the bias benchmarks, and it also transfers positively to adjacent video understanding tasks.

6 Analysis

6.1 Ablation of the training objective

Table 4 isolates the effect of each loss component on HMDB51. Several patterns are clear. First, using only the adversarial static loss drives SCUFO accuracy to 0.00% but leaves SCUBA accuracy much lower than the full model (42.29% versus 53.53%), suggesting that adversarial pressure alone is too blunt. Second, entropy maximization alone already helps considerably, improving contrasted accuracy from 27.84% to 44.30%. Third, the gradient penalty on its own provides limited benefit, but becomes useful in combination with the other terms. The full objective is the only configuration that simultaneously attains strong SCUBA accuracy, low SCUFO accuracy, and the best contrasted accuracy.

6.2 Which static frame should be repeated?

Table 5 evaluates the frame used to construct $\bar{x}$. Using the middle frame performs best overall, with 53.22% contrasted accuracy, slightly ahead of random sampling at 53.02%. First and last frames are notably weaker. A plausible explanation is that middle frames tend to better reflect the canonical action posture while being less affected by transition frames at the clip boundaries. This supports the design choice used in our default training configuration.

6.3 Qualitative evidence of shortcut reliance

Figure 3 shows examples where the baseline model predicts actions from spatial context rather than motion. In these cases, the scene or subject appearance is sufficiently correlated with the class that the model commits to the wrong label. These examples are not a substitute for quantitative analysis, but they are consistent with the benchmark results: the failure mode is not random error, but systematic over-reliance on static evidence.

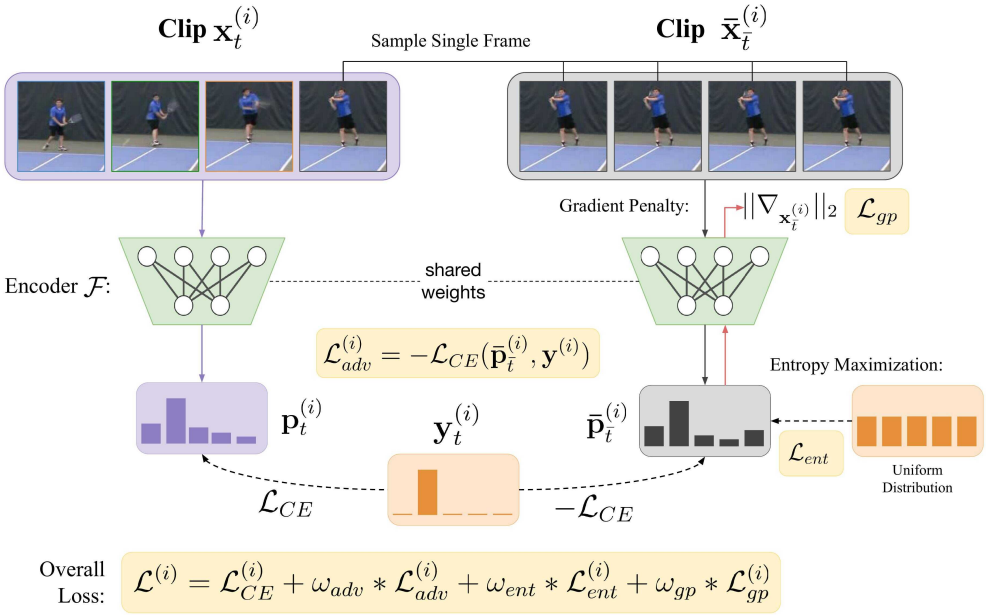


Figure 1: Overview of ALBAR. Given an input clip $\boldsymbol{x}$, we sample one frame and tile it across time to construct a static clip $\bar{\boldsymbol{x}}$. The classifier is trained to predict the action on $\boldsymbol{x}$ while becoming uninformative on $\bar{\boldsymbol{x}}$ through adversarial and entropy-based objectives.

Table 1: Comparison results on IID and OOD test sets of various debiasing methods on HMDB51. All experiments use Swin-T pretrained using Kinetics-400. The light blue column highlights the contrasted accuracy.

Augmentation or Debiasing	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
None	73.92	43.93	20.46	36.58	27.84
$\text{Mixup}_{\text{ICLR'18}}$	74.58	43.10	21.17	36.62	26.09
$\text{VideoMix}_{\text{arXiv'20}}$	73.31	39.39	20.44	32.68	23.13
$\text{SDN}_{\text{NeurIPS'19}}$	74.66	40.02	20.22	34.87	22.88
$\text{BE}_{\text{CVPR'21}}$	74.31	43.56	19.96	35.99	27.84
$\text{ActorCutMix}_{\text{CVIU'23}}$	74.05	46.79	22.07	36.97	28.12
$\text{FAME}_{\text{CVPR'22}}$	73.79	51.40	26.92	39.61	29.66
$\text{StillMix}_{\text{ICCV'23}}$	**74.82**	51.81	13.39	47.38	40.28
Ours	72.81	**53.53** ↑21.9%	**1.50** ↓92.7%	**48.13** ↑31.4%	**53.22** ↑91.2%
Ours w/ StillMix aug.	74.31	**54.24**	**1.35**	47.07	**53.68**

Table 2: Results on IID and OOD test sets of various debiasing methods on UCF101 using our proposed updated protocol. The light blue column highlights the contrasted accuracy. All experiments use Swin-T pretrained using Kinetics-400.

Augmentation or Debiasing	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
None	95.51	18.78	1.50	49.36	17.53
StillMix [ICCV'23]	**96.14**	24.68	0.36	**55.03**	24.40
Ours	94.98	**26.25** ↑39.8%	**0.14** ↓90.7%	54.02 ↑9.4%	**26.23** ↑49.6%
Ours w/ StillMix aug.	94.71	**31.17**	0.16	**58.84**	**31.14**

Table 3: Evaluation of our debiased model across downstream anomaly detection (UCF_Crime) and temporal action localization (THUMOS14) tasks. Higher is better for all columns.

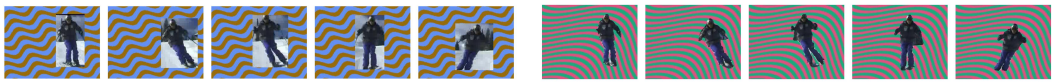
Method	**Action Recognition HMDB51 (OOD)**	**Anomaly Detection UCF_Crime**	**Temporal Action Localization THUMOS14**
Baseline	27.84	82.39	54.89
Ours	**53.22**	**84.91**	**55.20**

Table 4: Ablation to determine the effectiveness of each component loss objective. Each row keeps temporal supervision and toggles static regularizers.

	$\mathcal{L}_{CE}$	$\mathcal{L}_{adv}$	$\mathcal{L}_{ent}$	$\mathcal{L}_{gp}$	IID	OOD			
						Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
(a)	✓	✗	✗	✗	73.92	43.93	20.46	36.58	27.84
(b)	✓	✓	✗	✗	72.61	42.29	**0.00**	32.77	42.29
(c)	✓	✗	✓	✗	71.57	46.02	2.97	38.40	44.30
(d)	✓	✗	✗	✓	72.09	41.13	14.69	35.16	29.24
(e)	✓	✓	✓	✗	72.22	46.81	0.49	41.13	46.72
(f)	✓	✓	✗	✓	71.70	45.92	**0.00**	38.52	45.92
(g)	✓	✗	✓	✓	**73.86**	49.14	9.74	42.89	41.11
(h)	✓	✓	✓	✓	72.81	**53.53**	1.50	**48.13**	**53.22**

Table 5: Ablation to evaluate static frame sampling strategy.

Frame Sampling Strategy	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
Random	73.20	53.22	0.42	49.84	53.02
First	72.75	50.92	0.18	45.59	50.91
Middle	72.81	53.53	1.50	48.13	**53.22**
Last	72.68	49.49	0.40	42.42	49.40



(a) Earlier protocol with residual snow leakage. (b) Updated protocol with tighter segmentation masks.

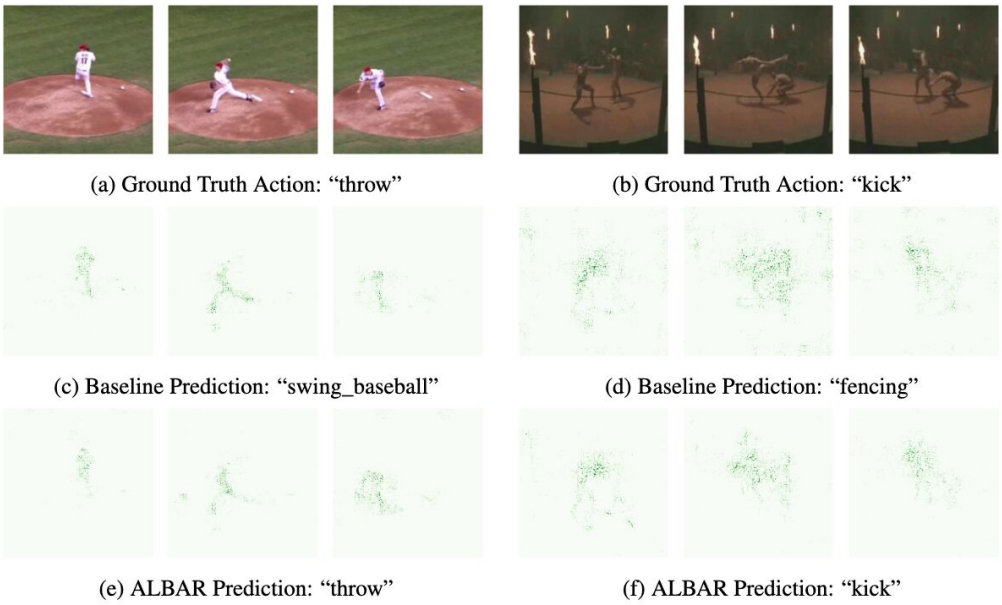
Figure 2: Why benchmark cleanup matters. In UCF101-SCUBA-Sinusoid, coarse compositing can leave class-correlated background cues visible around the actor. Our updated protocol uses tighter segmentation masks to better isolate the intended background intervention.

6.4 What the benchmark tells us

The updated UCF101 results highlight a broader point about evaluation. A debiasing benchmark must be adversarial not only to the classifier, but also to the protocol designer. If residual background cues remain in a supposedly debiased SCUBA clip, then the measured OOD performance can overstate real robustness. The metric suite is intentionally diagnostic: Avg SCUBA measures tolerance to background shift, Avg SCUFO measures residual static predictiveness, and contrasted accuracy rewards the desired asymmetry between them. This distinction matters empirically because a method can raise SCUBA accuracy while still extracting enough appearance information to remain correct on SCUFO, which would look better under a single OOD metric than it should. Our tighter compositing procedure does not solve every evaluation issue, but it materially improves the fidelity of the intervention by bringing the benchmark closer to its intended causal question: does the classifier still succeed when background bias changes and motion remains the only reliable evidence?

7 Conclusion

We presented two connected contributions for robust action recognition. The first is ALBAR, a simple training objective that discourages a classifier from encoding action labels in motionless clips by combining static adversarial supervision, entropy maximization, and gradient regularization. The second is a leakage-reduced UCF101-SCUBA/SCUFO benchmark that sharpens evaluation of background and foreground static bias through tighter segmentation-based compositing. Across HMDB51 and the updated UCF101 protocol, ALBAR achieves the best contrasted accuracy among the compared methods, and it combines effectively with StillMix for further gains. The downstream transfer results on UCF_Crime and THUMOS14 suggest that debiasing toward motion-centric representations can improve more than the benchmark itself. Together, the method and benchmark provide a concrete step toward action recognition models that rely less on spurious spatial shortcuts and more on the evidence that should matter: motion.



(a) Ground Truth Action: “throw”

(b) Ground Truth Action: “kick”

(c) Baseline Prediction: “swing_baseball”

(d) Baseline Prediction: “fencing”

(e) ALBAR Prediction: “throw”

(f) ALBAR Prediction: “kick”

Figure 3: Qualitative examples from HMDB51 where the baseline model chooses an incorrect label due to static spatial context. These failure cases align with the motivation for explicitly suppressing confidence on motionless clips.

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